Reset Early, Reset Often, Eliminate Models

How aggressively resetting connection tableaux can saturate the SAT solver and find an unsatisfiability proof faster.

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A traveller meets a husband and wife. He knows that either both of them are *knights*, who always tell the truth, or *knaves*, who always lie.

The husband makes a statement, and the traveller is given three clues.

1. If the husband's statement is false, then his wife is a knight.
2. If the wife is a knight, then the husband's statement is true.
3. If the husband is a knight, then his wife is a knight

PUZ020-1.p

$$a(\eta) \qquad \neg b(\eta)$$

$$\neg a(A) \vee \neg c\big(\phi(A)\big) \vee b(A)$$

$$c\big(\phi(\eta)\big) \vee b(\gamma) \qquad \neg b(\gamma) \vee c\big(\phi(\eta)\big)$$

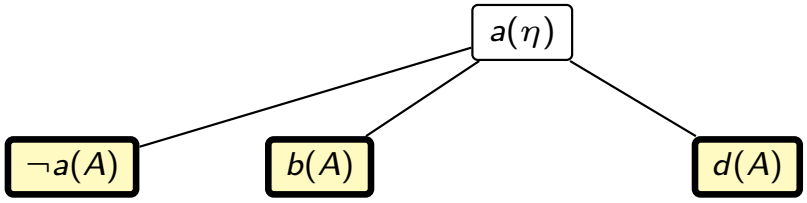
$a(A)$: A is a person.
$b(A)$: A is a knight.
$c(B)$: B is a true statement.
$d(A)$: A is a knave.

$e(A, B)$: A says statement B.
$\phi(A)$: statement ϕ made by A.
η: the husband.
γ: the wife.

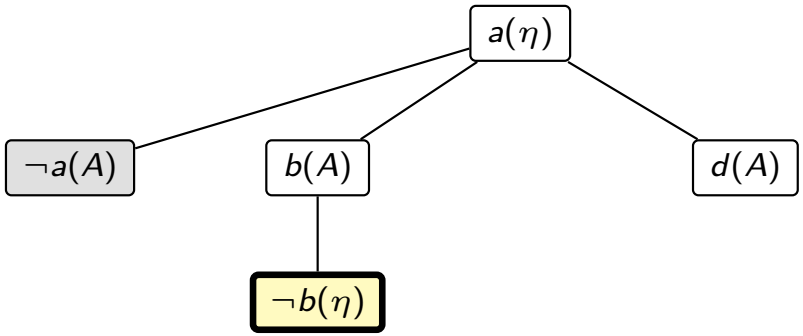
Start: choose the husband fact.

$a(\eta)$

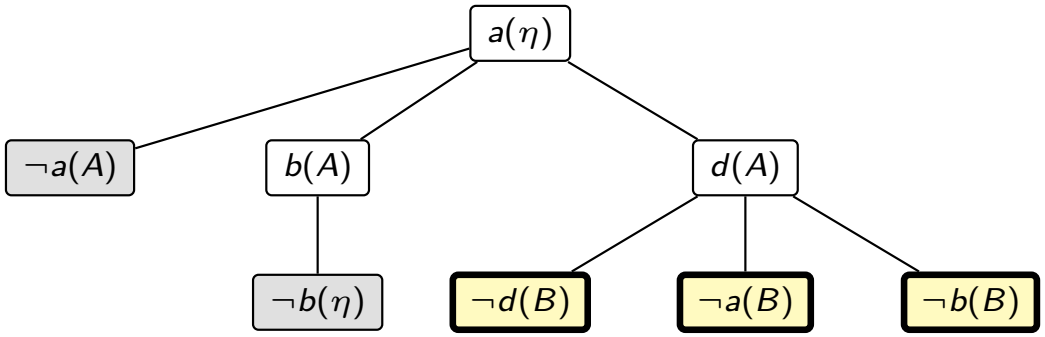
Extension: every person is a knight or a knave; $\neg a(A)$ closes.



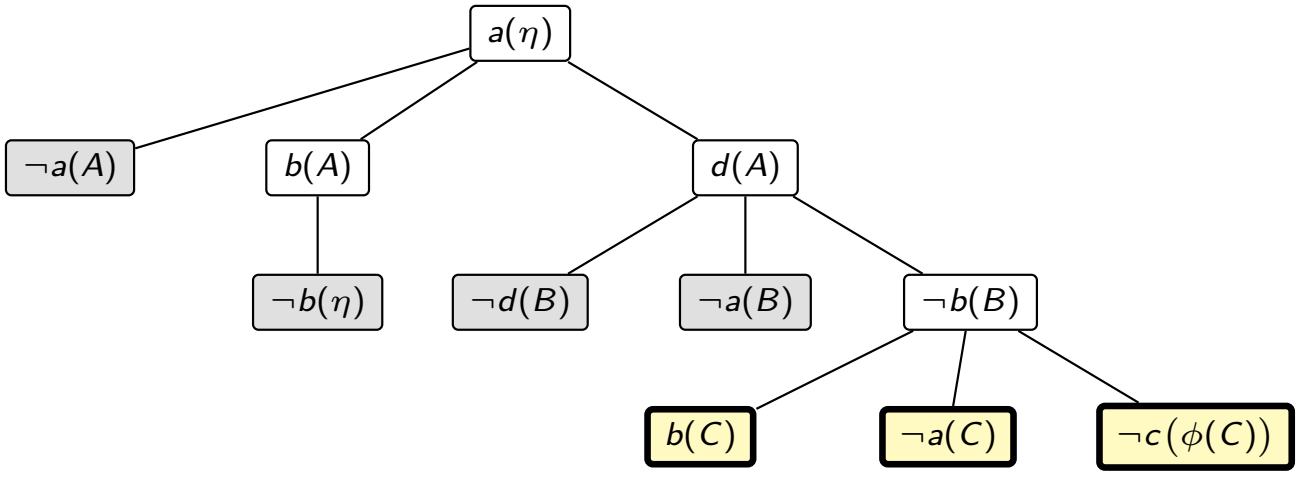
The knight branch closes against the negated conjecture.



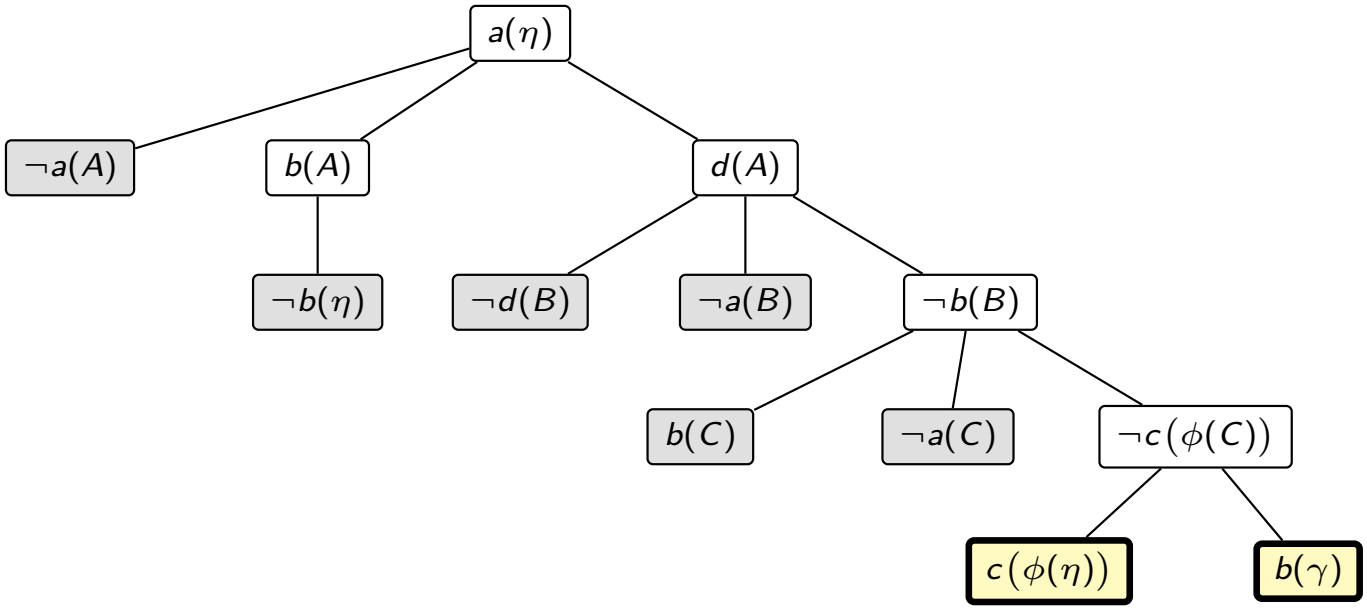
The knave branch expands using exclusivity of knights and knaves.



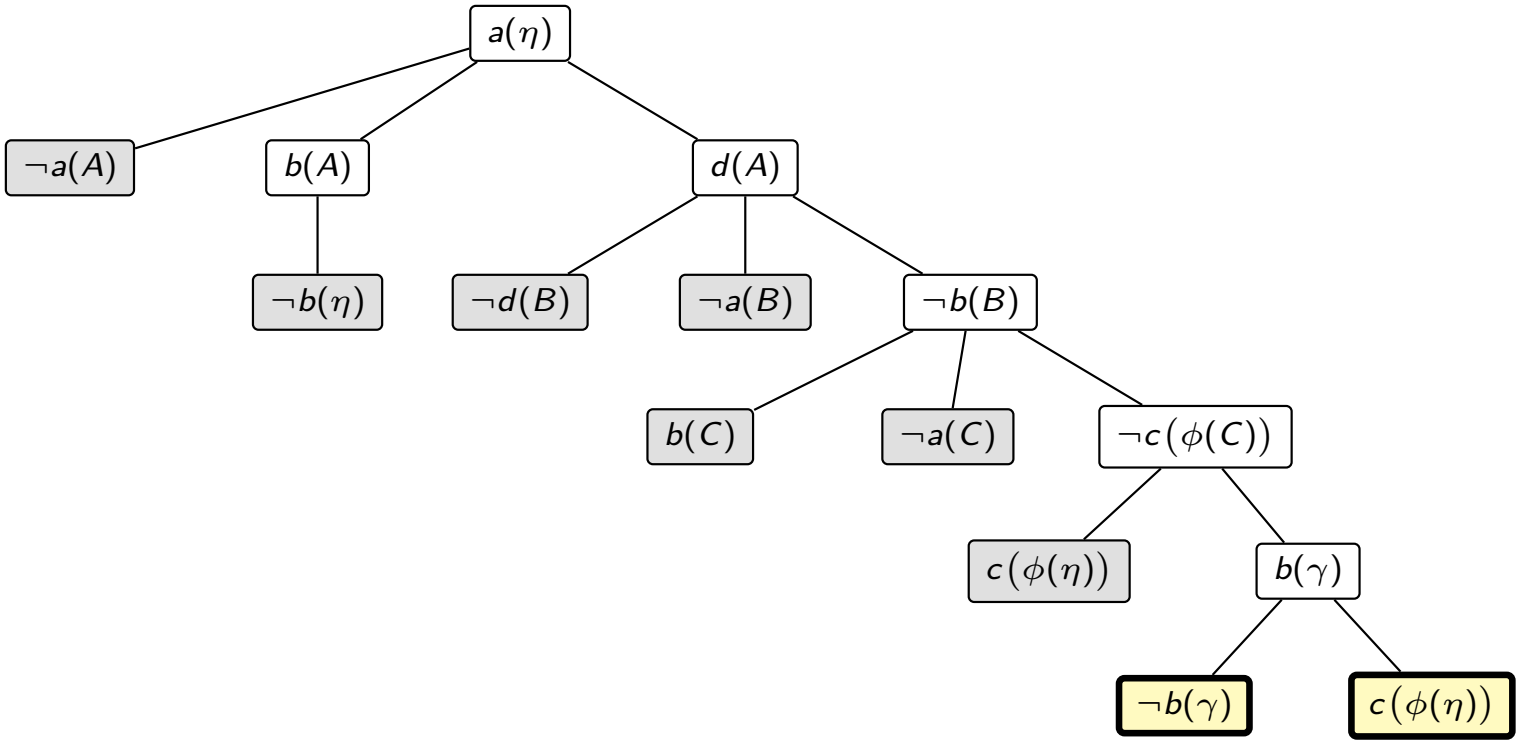
A second expansion relates truthful statements to knighthood.



If the statement were false, the wife would be a knight.



The wife would make the statement true: every branch closes.

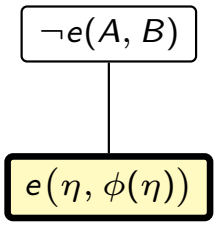


successful tableau — found after 2,563 search steps

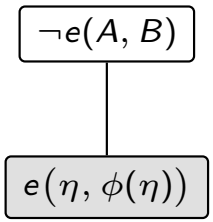
Start: open the husband-says branch.

$\neg e(A, B)$

It closes immediately with the statement fact.

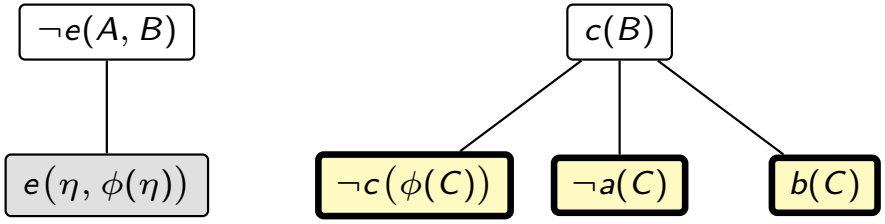


Start: now open the true-statement branch.

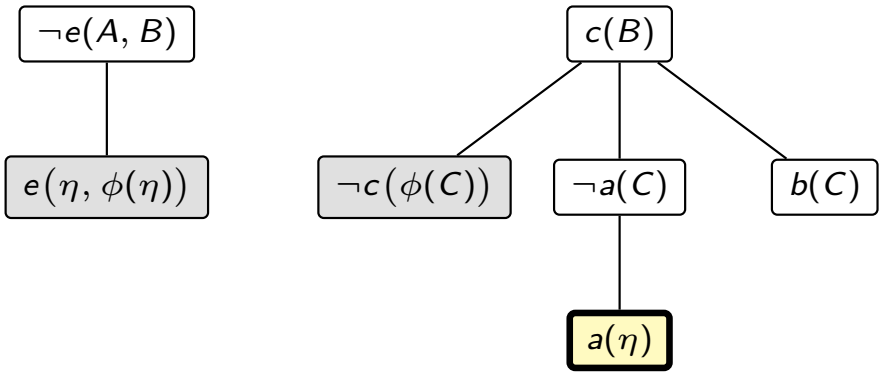


$c(B)$

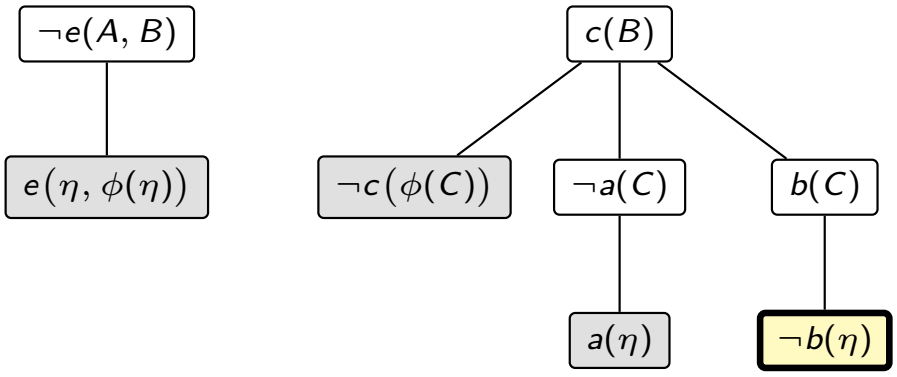
It connects to the truth/knight schema.



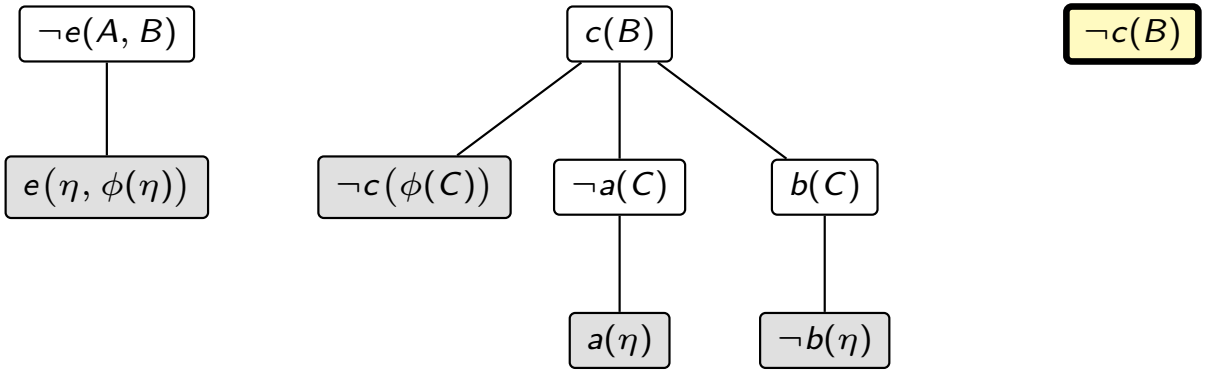
The person subgoal closes with the husband fact.



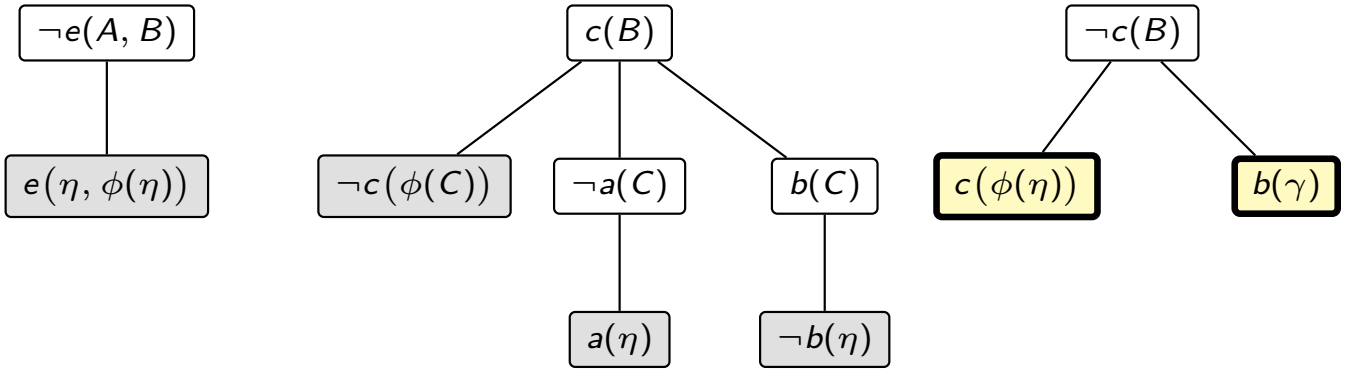
The knight subgoal closes against the conjecture.



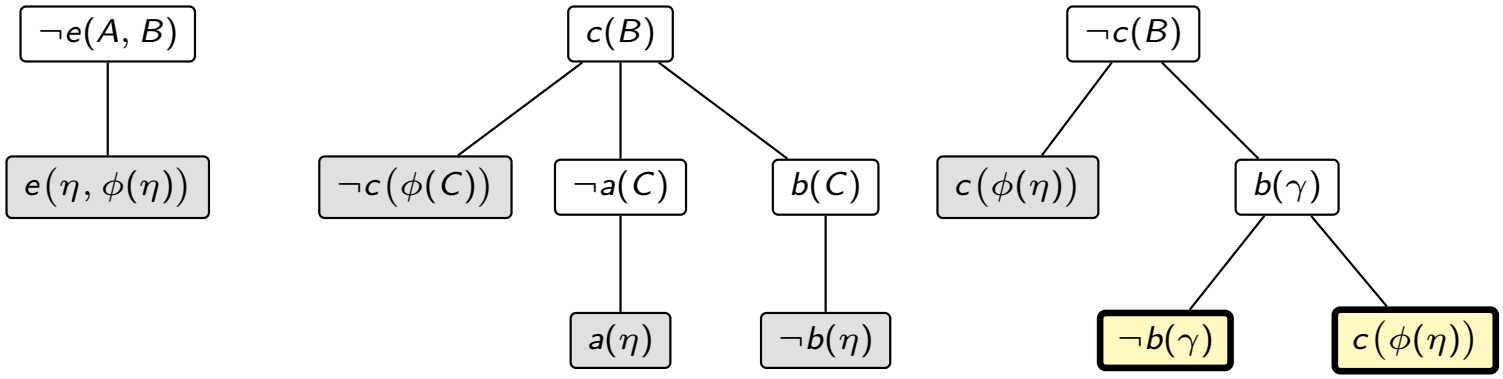
Start: finally open the false-statement branch.



A false statement forces the wife to be a knight.



That makes the statement true, closing the last branch.



successful tableau — found after 241 search steps

Start: begin with the husband fact.

$a(\eta)$

persistent SAT ground

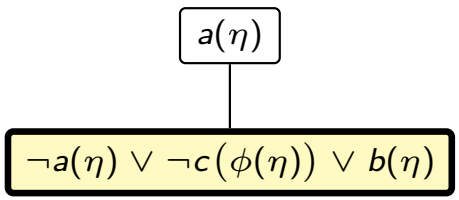
Ground that clause and add it to the persistent SAT state.

$a(\eta)$

persistent SAT ground

$a(\eta)$

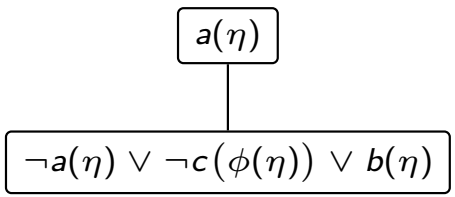
Extension: instantiate the truth/knight schema.



persistent SAT ground

$a(\eta)$

The new ground instance is retained by the SAT solver.

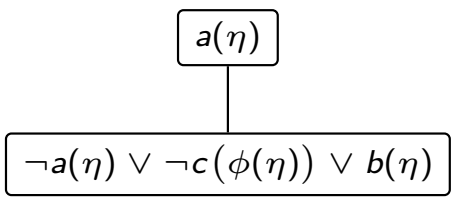


persistent SAT ground

$a(\eta)$

$\neg a(\eta) \vee \neg c(\phi(\eta)) \vee b(\eta)$

Try the negated conjecture as another start.



$\neg b(\eta)$

persistent SAT ground

$a(\eta)$ $\neg b(\eta)$ $\neg a(\eta) \vee \neg c(\phi(\eta)) \vee b(\eta)$

Try the clause saying a false statement makes the wife a knight.

$a(\eta)$

$\neg a(\eta) \vee \neg c(\phi(\eta)) \vee b(\eta)$

$\neg b(\eta)$

$c(\phi(\eta)) \vee b(\gamma)$

persistent SAT ground

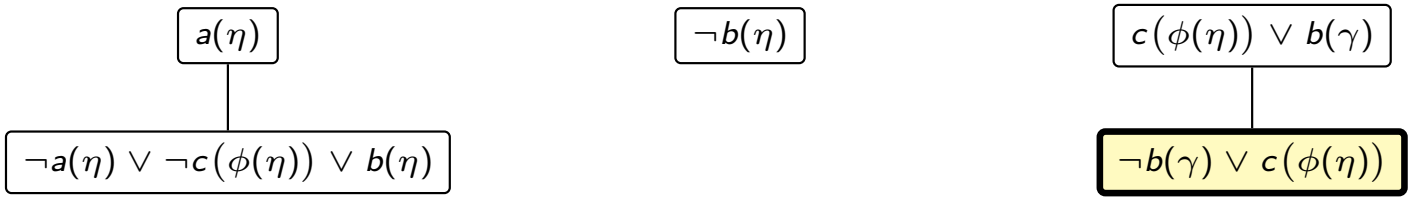
$a(\eta)$

$\neg b(\eta)$

$\neg a(\eta) \vee \neg c(\phi(\eta)) \vee b(\eta)$

$c(\phi(\eta)) \vee b(\gamma)$

Connect the wife branch; one more ground clause is learned.

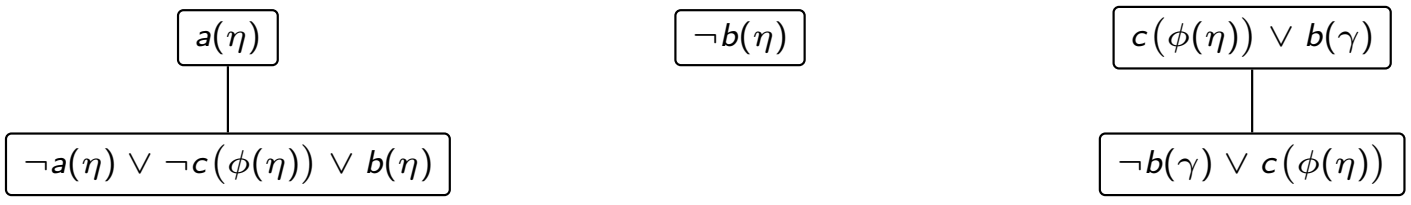


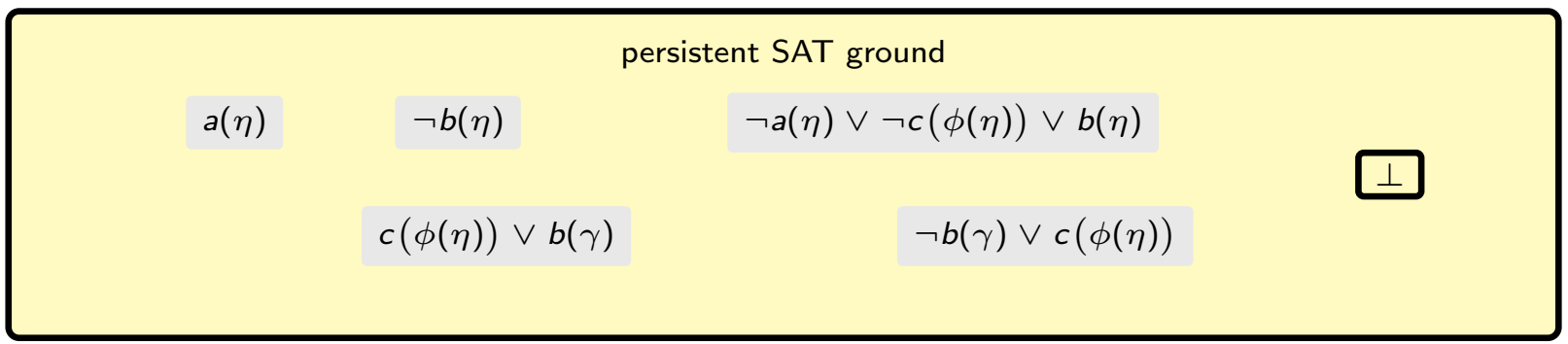
persistent SAT ground

$a(\eta)$ $\neg b(\eta)$ $\neg a(\eta) \vee \neg c(\phi(\eta)) \vee b(\eta)$

$c(\phi(\eta)) \vee b(\gamma)$ $\neg b(\gamma) \vee c(\phi(\eta))$

The accumulated ground is now UNSAT.





The SAT model prefers the wife/truth clause as a start.

$c(\phi(\eta)) \vee b(\gamma)$

persistent SAT ground

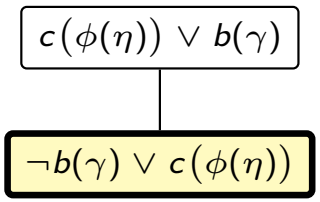
Its ground instance enters the persistent SAT state.

$c(\phi(\eta)) \vee b(\gamma)$

persistent SAT ground

$c(\phi(\eta)) \vee b(\gamma)$

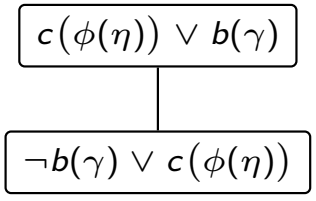
The selected wife literal connects to the converse implication.



persistent SAT ground

$c(\phi(\eta)) \vee b(\gamma)$

That second ground clause is retained.

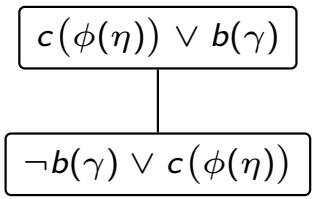


persistent SAT ground

$c(\phi(\eta)) \vee b(\gamma)$

$\neg b(\gamma) \vee c(\phi(\eta))$

The model next selects the husband fact.



$a(\eta)$

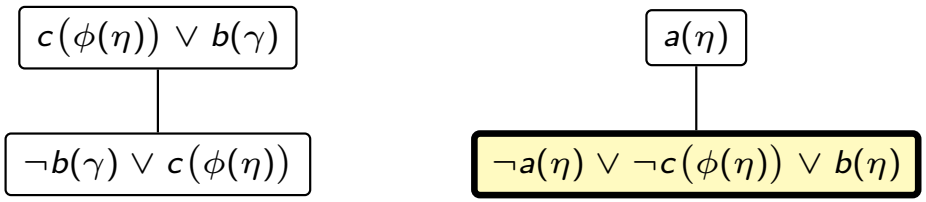
persistent SAT ground

$a(\eta)$

$c(\phi(\eta)) \vee b(\gamma)$

$\neg b(\gamma) \vee c(\phi(\eta))$

An extension adds the decisive truth/knight instance.



persistent SAT ground

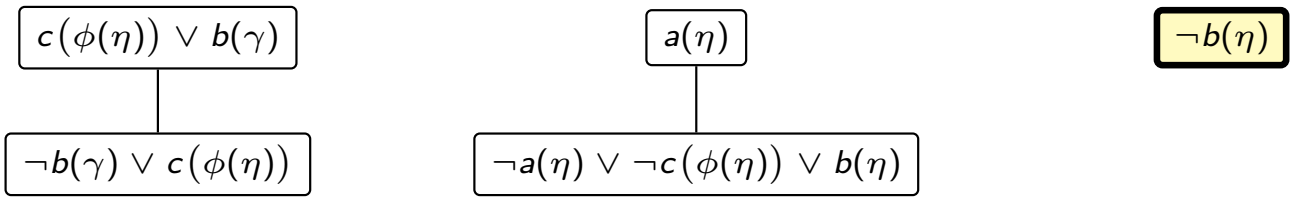
$a(\eta)$

$\neg a(\eta) \vee \neg c(\phi(\eta)) \vee b(\eta)$

$c(\phi(\eta)) \vee b(\gamma)$

$\neg b(\gamma) \vee c(\phi(\eta))$

The negated conjecture supplies the final missing clause.

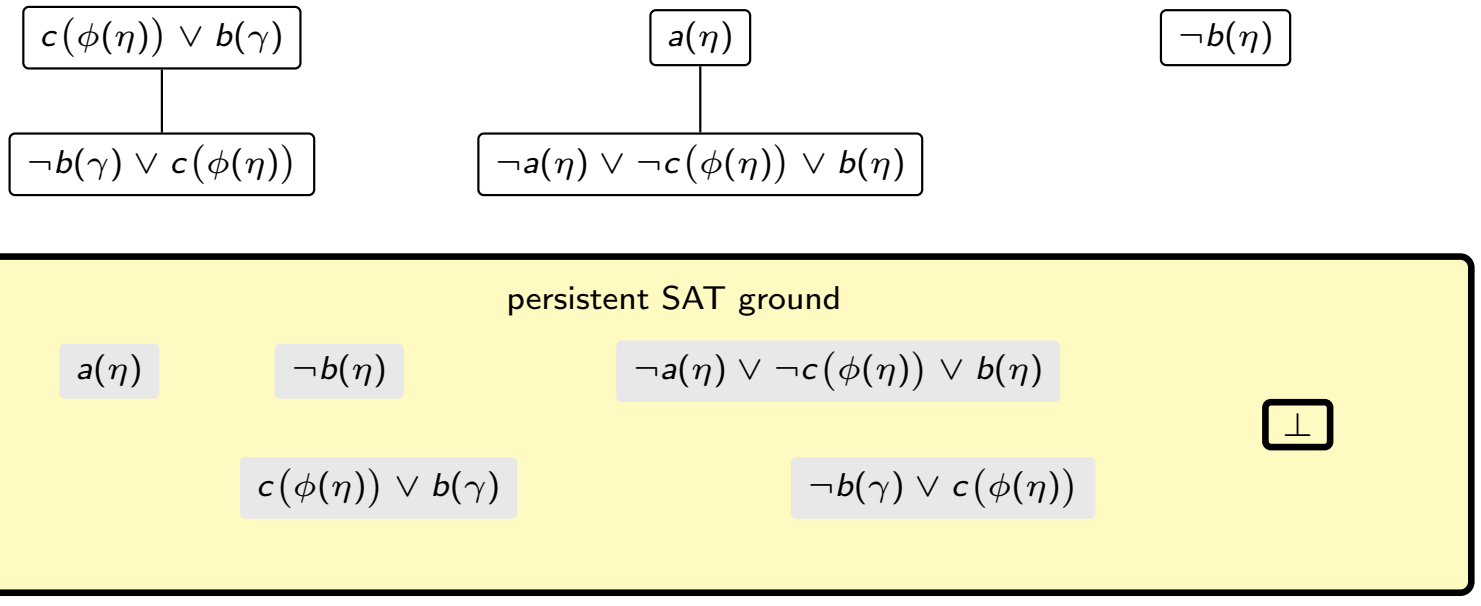


persistent SAT ground

$a(\eta)$ $\neg b(\eta)$ $\neg a(\eta) \vee \neg c(\phi(\eta)) \vee b(\eta)$

$c(\phi(\eta)) \vee b(\gamma)$ $\neg b(\gamma) \vee c(\phi(\eta))$

The accumulated ground is now UNSAT.

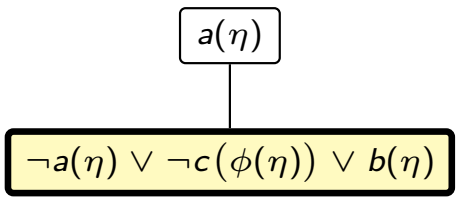


First attempt: start from the husband fact.

$a(\eta)$

persistent SAT ground

One extension contributes two useful ground clauses.

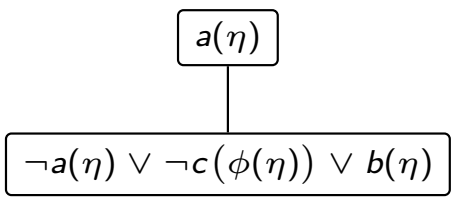


persistent SAT ground

$a(\eta)$

$\neg a(\eta) \vee \neg c(\phi(\eta)) \vee b(\eta)$

Reset: discard the tableau and start from the wife/truth clause.



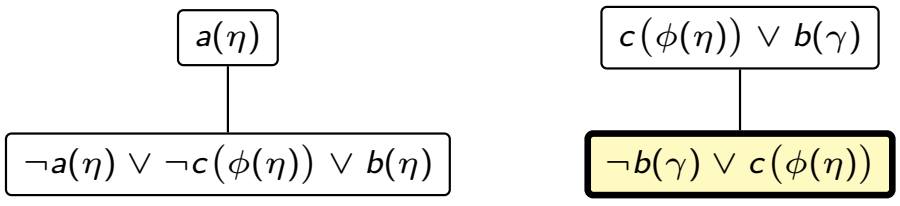
$c(\phi(\eta)) \vee b(\gamma)$

persistent SAT ground

$a(\eta)$

$\neg a(\eta) \vee \neg c(\phi(\eta)) \vee b(\eta)$

Another short attempt contributes both wife clauses.



persistent SAT ground

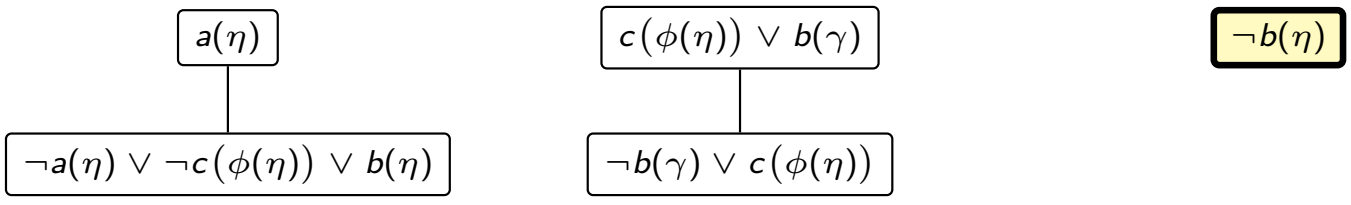
$a(\eta)$

$\neg a(\eta) \vee \neg c(\phi(\eta)) \vee b(\eta)$

$c(\phi(\eta)) \vee b(\gamma)$

$\neg b(\gamma) \vee c(\phi(\eta))$

Reset: the negated conjecture supplies the final clause.

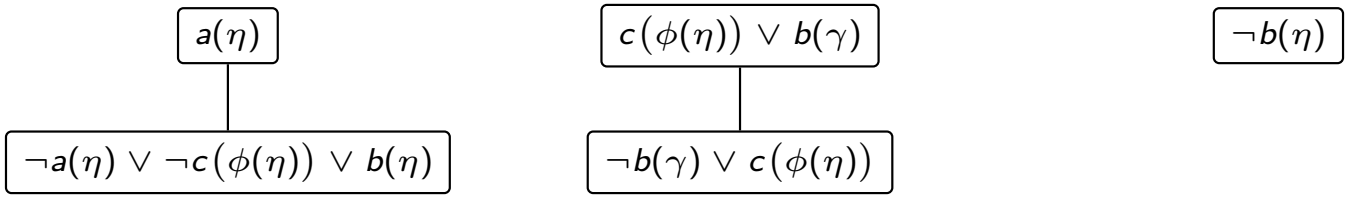


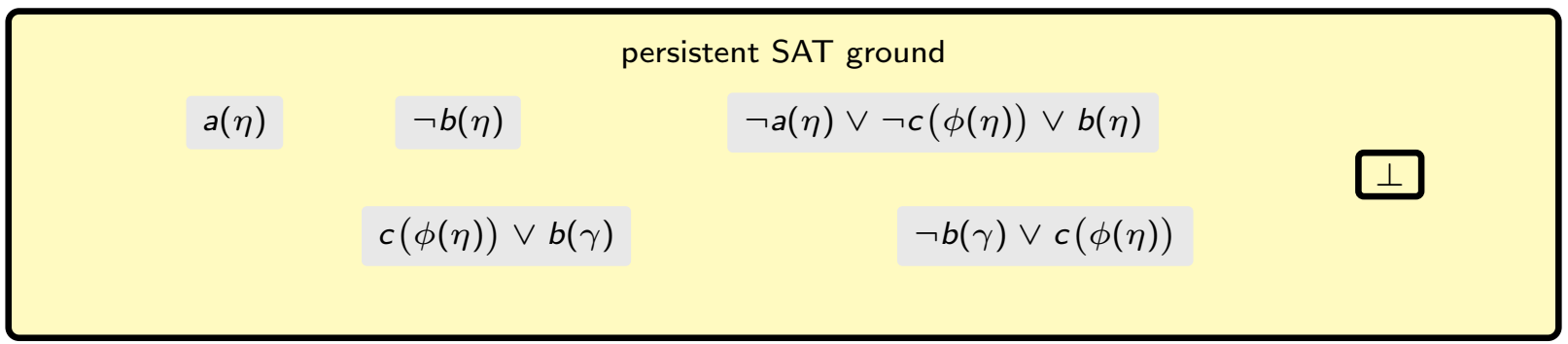
persistent SAT ground

$a(\eta)$ $\neg b(\eta)$ $\neg a(\eta) \vee \neg c(\phi(\eta)) \vee b(\eta)$

$c(\phi(\eta)) \vee b(\gamma)$ $\neg b(\gamma) \vee c(\phi(\eta))$

The tableau is disposable; the persistent ground is UNSAT.





Why is SATCoP so good?

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Further Improvements to SATCoP

What if we stop doing regular tableau guidance?

General Results

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References